

Challenges of Building a Non-Custodial Lightning App

ek

STACKER NEWS

October 3, 2025

Agenda

- 1 Why Non-Custodial?
- 2 Custodial vs Non-Custodial Architecture
- 3 User Experience
- 4 Challenges

Why Non-Custodial?

Wallet of Satoshi stopped serving US customers

NEWS 1 MIN READ 24 NOV 2023

Wallet of Satoshi to Stop Serving Customers in the United States

"We've made the difficult decision to remove our app from the U.S. Apple and Google app stores, and will not serve U.S. customers going forward," announced Wallet of Satoshi.



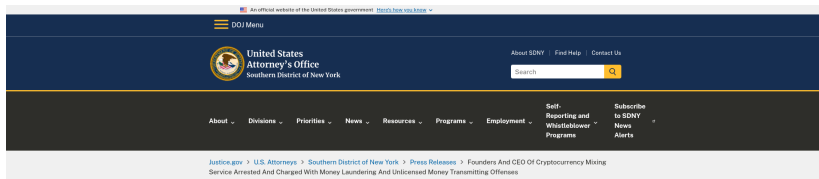
- "For our existing users in the USA, rest assured that you have full access to your Bitcoin funds. You can seamlessly withdraw and transfer them to another wallet," was stated in the post.

Wallet of Satoshi is a custodial, closed-source Bitcoin and Lightning wallet and service provider based in Australia. The wallet became popular due to its ease of use and frictionless onboarding experience.

- "This decision doesn't come lightly. Our commitment to providing a secure, user-friendly, and compliant platform globally is unwavering. Our top priority is the safety and interests of our customers and our company."

<https://www.nobsbitcoin.com/wallet-of-satoshi-stops-serving-us-customers/>

Samourai Devs prosecuted



PRESS RELEASE

Founders And CEO Of Cryptocurrency Mixing Service Arrested And Charged With Money Laundering And Unlicensed Money Transmitting Offenses

Wednesday, April 24, 2024

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For Immediate Release

U.S. Attorney's Office, Southern District of New York

Keonne Rodriguez and William Lonergan Hill Are Charged with Operating Samourai Wallet, an Unlicensed Money Transmitting Business That Executed Over \$2 Billion in Unlawful Transactions and Laundered Over \$100 Million in Criminal Proceeds

Damian Williams, the United States Attorney for the Southern District of New York; Thomas Fattorusso, the Special Agent in Charge of the New York Field Office of the Internal Revenue Service, Criminal Investigation ("IRS-CI"); and James Smith, the Assistant Director in Charge of the New York Field Office of the Federal Bureau of Investigation ("FBI"), announced today the unsealing of an indictment charging KEONNE RODRIGUEZ, the Chief Executive Officer and a co-founder of Samourai Wallet ("Samourai"), and WILLIAM LONERGAN HILL, the Chief Technology Officer and also a co-founder of Samourai, with conspiracy to

<https://www.justice.gov/usao-sdny/pr/founders-and-ceo-cryptocurrency-mixing-service-arrested-and-charged-money-laundering>

Phoenix Wallet stopped serving US customers

NEWS 1 MIN READ 25 APR 2024

Phoenix Wallet to Stop Serving US Customers on May 3rd

Phoenix wallet users in the United States are encouraged to close their channels and empty their wallets prior to May 3, 2024.



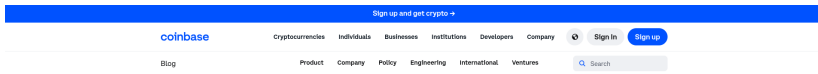
- "On May 3rd, 2024, Phoenix Wallet will be removed from US app stores," announced the project.

"Recent announcements from US authorities cast a doubt on whether self-custodial wallet providers, Lightning service providers, or even Lightning nodes could be considered Money Services Businesses and be regulated as such. We are removing Phoenix Wallet from US app stores and will be assessing other potential impacts to our operations," said ACINQ, the developer behind Phoenix Wallet.

The project developer added that users from the US should empty their wallet by going to:

<https://www.nobsbitcoin.com/phoenix-wallet-to-be-removed-from-us-app-stores-on-may-3/>

New York's BitLicense . . .



Coinbase Obtains the BitLicense

By Juan Suarez | [Coinbase](#), January 12, 2017, 1 min read time

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We are excited to announce that Coinbase has obtained the BitLicense from the New York Department of Financial Services. The BitLicense, which authorizes Coinbase's continuing virtual currency business operations in the state of New York, is an important validation of our highest priority: to operate the most [secure](#) and [compliant](#) digital currency exchange in the world. We believe the long-term success of virtual currency and blockchain technologies will require productive partnerships between industry and government, and we will continue to make heavy investments in the security and integrity of our operations as we expand our business in New York and around the world.



<https://www.coinbase.com/en-de/blog/coinbase-obtains-the-bitlicense>

... is not enough for the Lightning Network

UPDATE

Coinbase Integrates Bitcoin Lightning Network for US Exchange Users

By Vivek Sen April 30, 2024



Bitcoin and crypto exchange Coinbase has integrated Bitcoin's Lightning Network, enabling faster and cheaper BTC transactions for its users in the United States (with the exception of New York).

As the largest Bitcoin exchange globally by users, trading volume, and custody of assets, Coinbase's adoption of Lightning is a major win for the Bitcoin scaling solution. Lightning allows near-instant Bitcoin payments with significantly lower fees by moving transactions off-chain.

Coinbase has been working to implement Lightning support for years to further its mission of building a payments network costing pennies per transaction. This integration finally unlocks the power of instant and affordable Bitcoin payments for the exchange's massive US user base.

<https://bitcoinmagazine.com/business/coinbase-integrates-bitcoin-lightning-for-100-million-users>

Paying \$\$\$ for the privilege to:

- implement KYC
- monitor transactions
- report suspicious activity

“Soft KYC”

 **Activate Wallet**



Activating your wallet is easy!
We just need a few details below:

We said no.

... but still had to comply with OFAC sanctions ...

All U.S. persons must comply with OFAC sanctions, including all U.S. citizens and permanent residents regardless of where they are located, all individuals and entities within the United States, and all U.S. incorporated entities and their foreign branches.

<https://ofac.treasury.gov/faqs/11>

OFAC

```
api/resolvers/ofac.js
... @@ -0,0 +1,24 @@
1 + import { GraphQLError } from 'graphql'
2 +
3 + // this function makes america more secure apparently
4 + export default async function assertGofacYourself ({ models, headers }) {
5 +   const country = await goFACYourself({ models, headers })
6 +   if (!country) return
7 +
8 +   throw new GraphQLError(
9 +     `Your IP address is in ${country}. We cannot provide financial services to residents of ${country}.`,
10 +    { extensions: { code: 'FORBIDDEN' } })
11 + }
```

<https://github.com/stackernews/stacker.news/commit/a5e50821b7e0dba0400a5f8ada6d3741a92de9d3>

Custodial vs Non-Custodial Architecture

Custodial Architecture

- “zaps” are database transactions
- only deposits and withdrawals are lightning payments
- lightning node can go down and most users wouldn't notice

Is this really a lightning app?

Non-Custodial Architecture

- (almost) every payment is over lightning
- senders now have to pay routing fees¹
- zaps need to be forwarded to receiver

How to never take custody of funds while forwarding?

¹if they don't have a channel with SN

↔ Inproxy.org

about
api
contact

Random relay

Inbc...

Wrap

more options ◀

dx7pn6ehykq6cadce4bjbxn5tf64z7e3fufpxgxce7n4f5eja476cpyd.onion

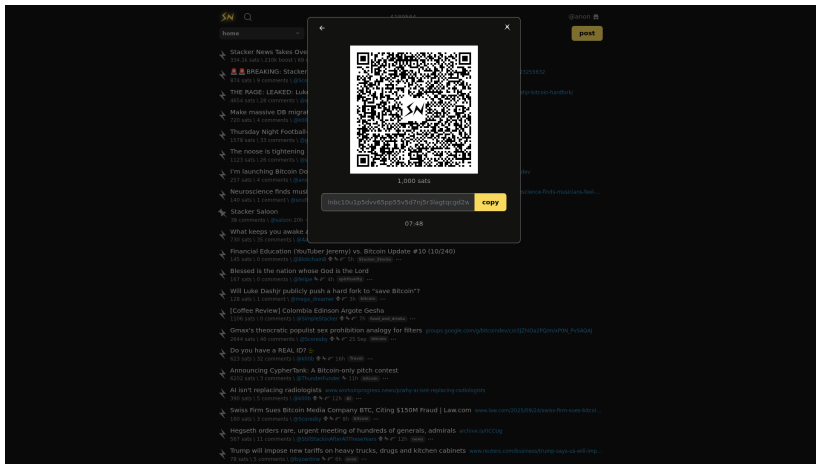
Demo

The End.

I wish ...

User experience

QR codes are not good UX

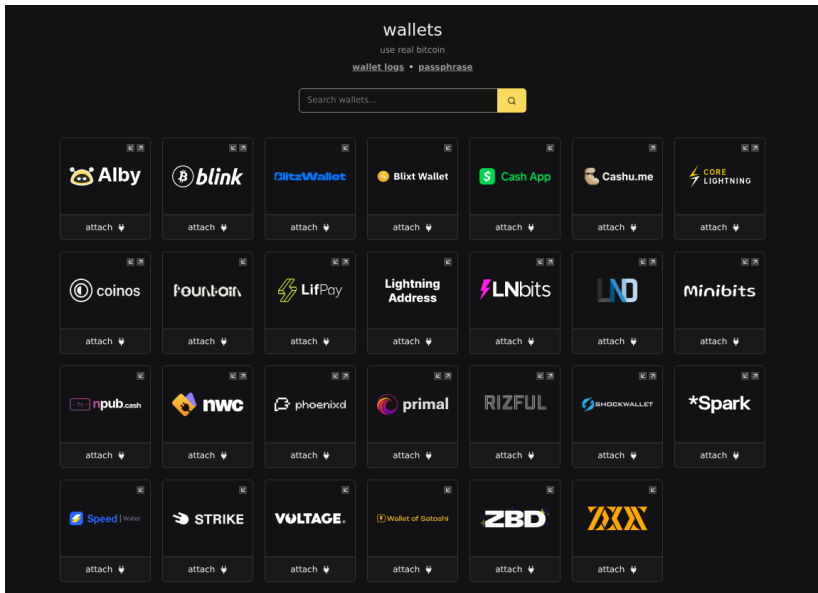


QR codes are not good UX

- desktop vs mobile
- everything about a payment can be slow

What if the payment fails?

Wallets



Demo

Payment UX

- optimistic payments
- retry payments in the background
- notify user if a payment failed

should *feel* like custodial

Payment retries

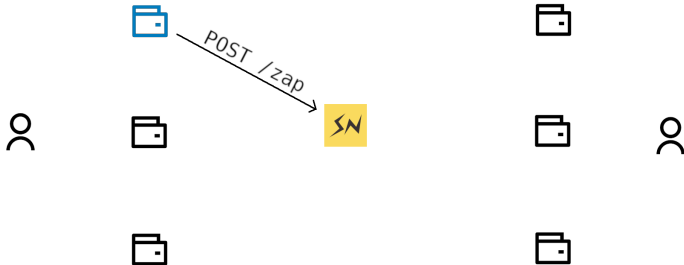
- payments are attempted three times
- users can attach multiple wallets as fallbacks
- did sender or receiver fail?

Never reuse a lightning invoice!

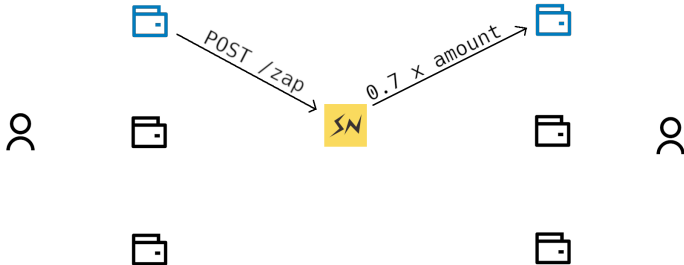
P2P zaps



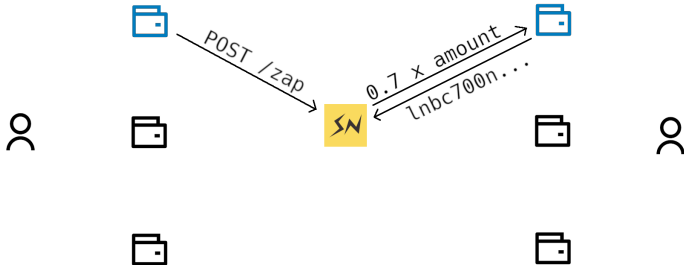
P2P zaps



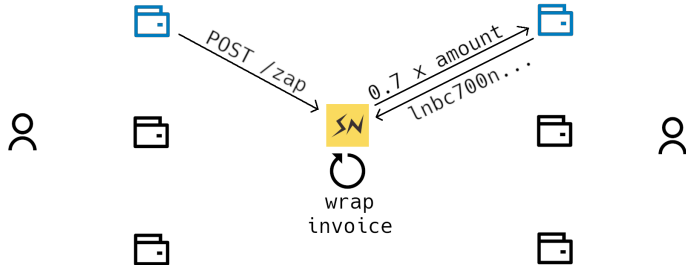
P2P zaps



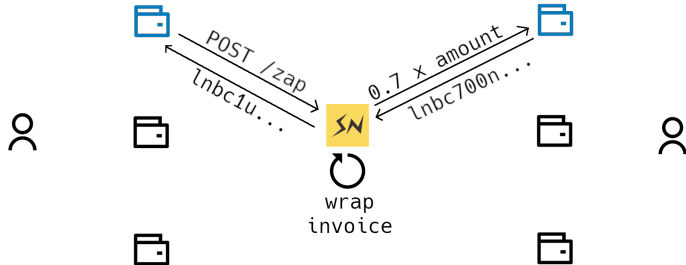
P2P zaps



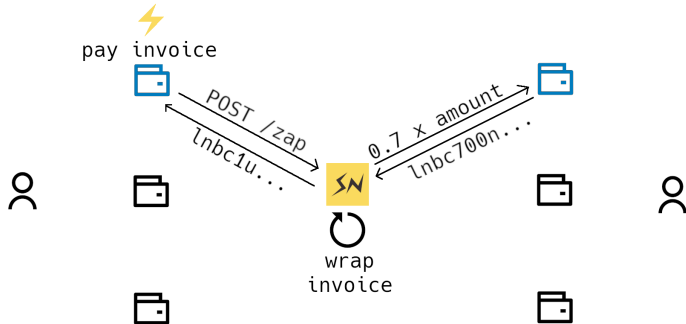
P2P zaps



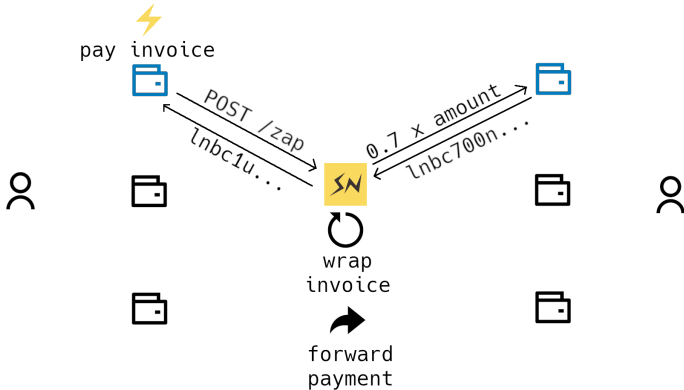
P2P zaps



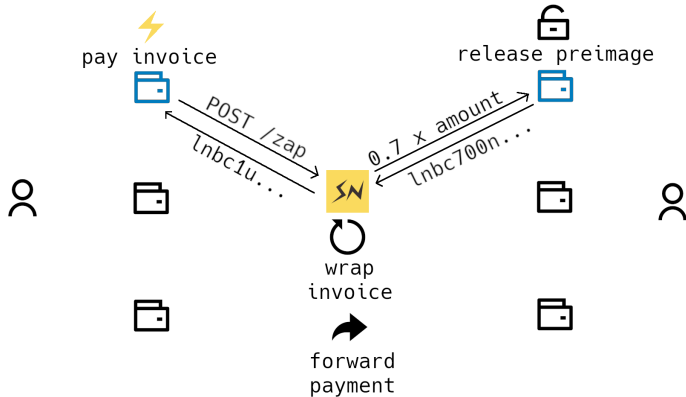
P2P zaps



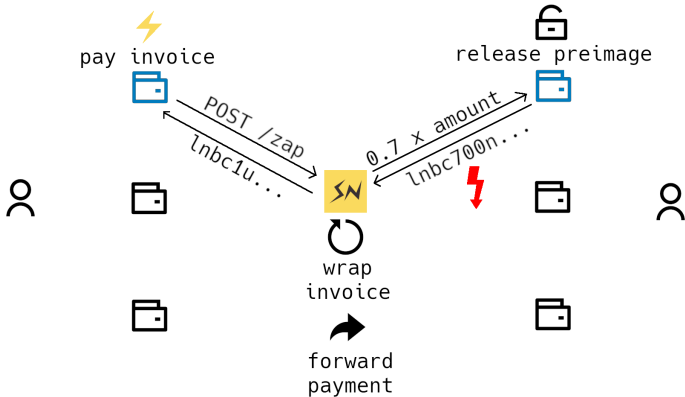
P2P zaps



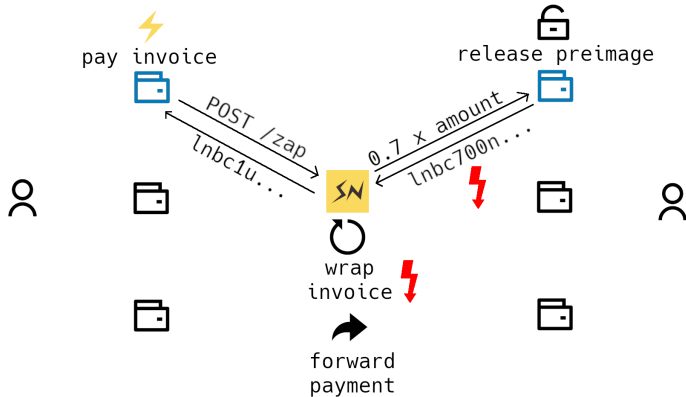
P2P zaps



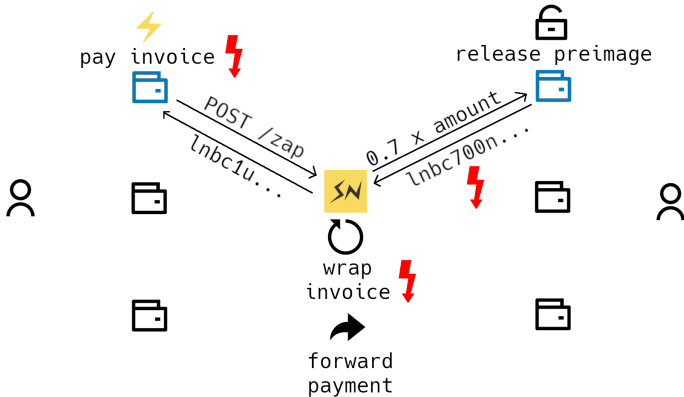
P2P zaps



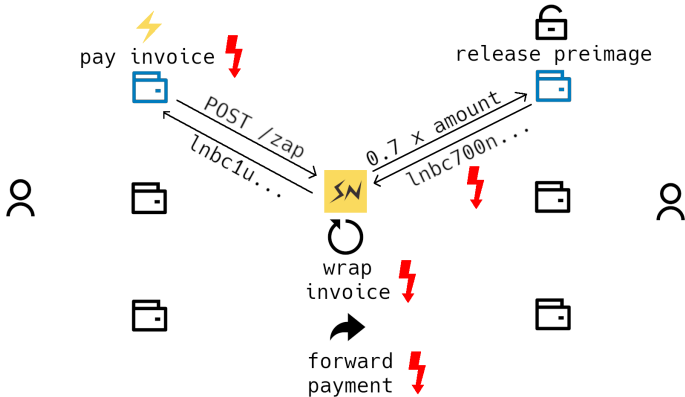
P2P zaps



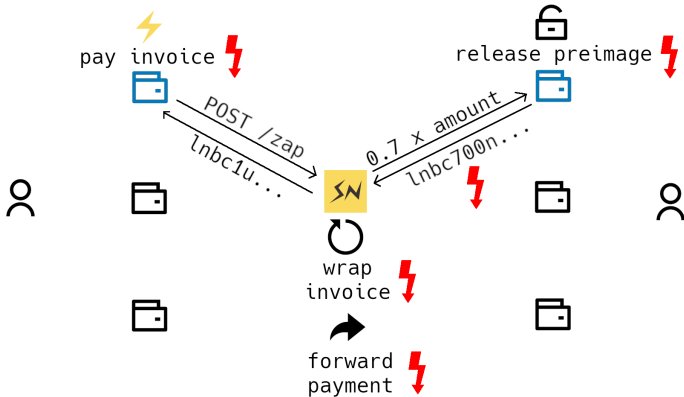
P2P zaps



P2P zaps



P2P zaps



Summary:

Lightning on the Application layer

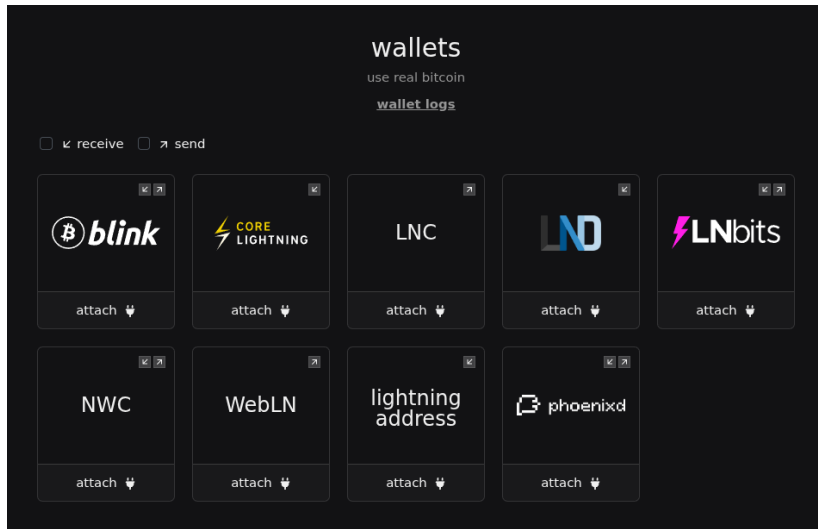
Reminder to show notifications

Challenges

Challenges

- zaps must include forwarding fee
- wallets vs protocols

Challenges



Challenges

- share wallets between multiple devices
- encrypted send credentials vs plaintext receive credentials
- how to test send credentials?

Challenges

- user education
- high availability lightning node
- liquidity issues
- forwarding to multiple receivers

What if somebody has no lightning wallet?

Cowboy Credits

Cowboy Credits exist to smooth over UX issues:

- new users do not have a wallet
- receiving sats can fail for many reasons
- some users prefer UX over sats

CCs are non-withdrawable

Cowboy Credits

Also:

- used first to pay SN instead of sats
- backup for lightning network issues
- 30% of a zap go to territory founders and rewards **as sats**

Half-life of a CC is two zaps

The End



special thanks to
@plebpoet

